

Aim / Objective

To demonstrate method overriding and runtime polymorphism in Java by calculating different rates of interest for different banks.

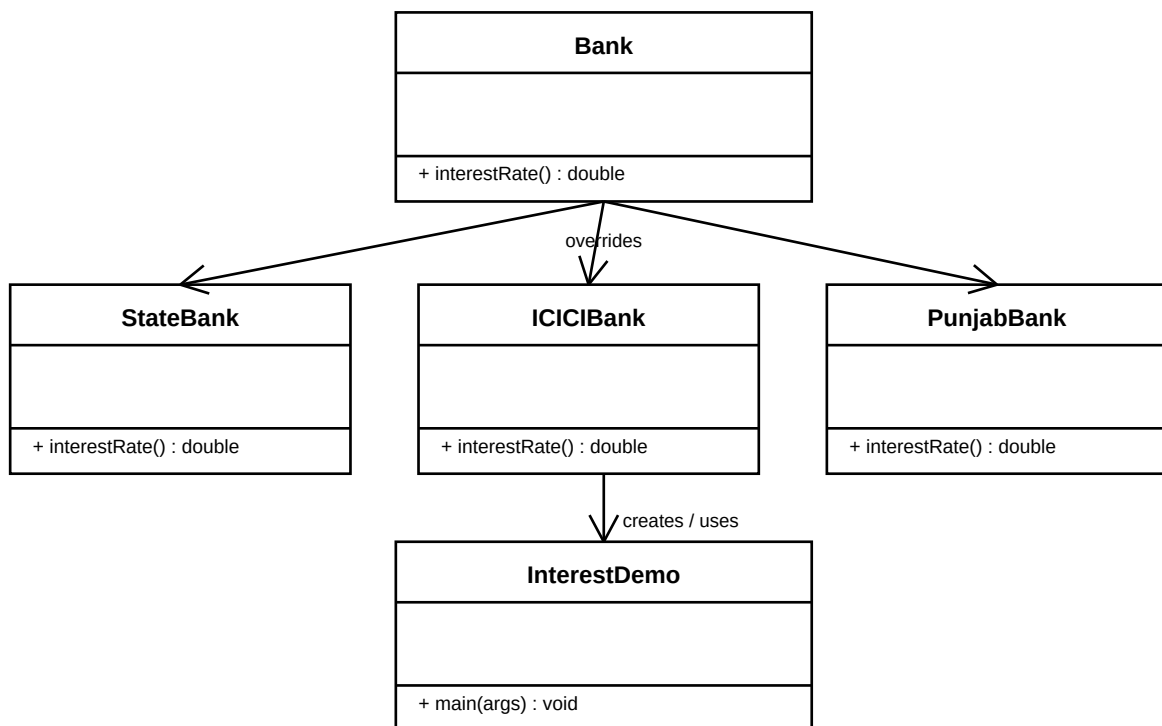
System Requirements

Java Development Kit (JDK) and any Java IDE such as Eclipse or IntelliJ IDEA.

Algorithm / Logic Description

1. Create a Bank class with an interestRate() method returning the base interest rate.
2. Create StateBank, ICICIBank, and PunjabBank classes by extending Bank.
3. Override the interestRate() method in each child class with its respective rate.
4. Read the bank name from the user using Scanner.
5. Create a Bank reference and assign the corresponding child-class object based on the entered bank name.
6. Call interestRate() through the Bank reference to demonstrate runtime polymorphism.
7. Display the selected bank's rate of interest.

UML Class Diagram



Source Code

```
package core_java;

import java.util.Scanner;

class Bank {

    double interestRate() {
        return 4.0;
    }
}
```

```

    }
}

class StateBank extends Bank {

    @Override
    double interestRate() {
        return 7.0;
    }
}

class ICICIBank extends Bank {

    @Override
    double interestRate() {
        return 6.5;
    }
}

class PunjabBank extends Bank {

    @Override
    double interestRate() {
        return 6.0;
    }
}

public class InterestDemo {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        System.out.print("Enter bank name: ");
        String name = input.nextLine();

        Bank bankObj;

        if (name.equalsIgnoreCase("RBI")) {

            bankObj = new Bank();

        } else if (name.equalsIgnoreCase("SBI")) {

            bankObj = new StateBank();

        } else if (name.equalsIgnoreCase("ICICI")) {

            bankObj = new ICICIBank();

        } else if (name.equalsIgnoreCase("PNB")) {

            bankObj = new PunjabBank();

        } else {

            System.out.println("Bank name is not available.");
            input.close();
            return;
        }

        double rate = bankObj.interestRate();

        System.out.println("Rate of Interest: " + rate + "%");

        input.close();
    }
}

```

Compilation and Execution Commands

```

javac InterestDemo.java
java InterestDemo

```

Output

```
PS C:\Users\Student> javac InterestDemo.java
PS C:\Users\Student> java InterestDemo
Enter bank name: SBI
Rate of Interest: 7.0%
PS C:\Users\Student>
```

Result

The program was implemented successfully to demonstrate method overriding and runtime polymorphism for different banks.